

mNGS

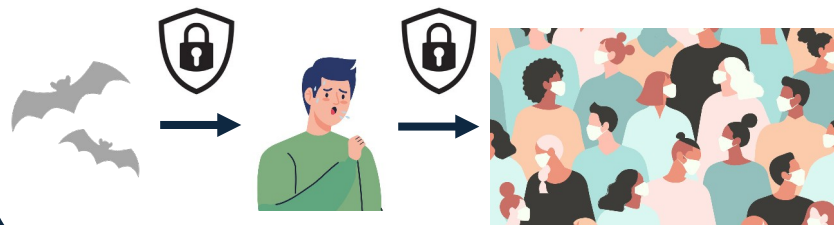
Study design and sample preparation

Use cases for mNGS

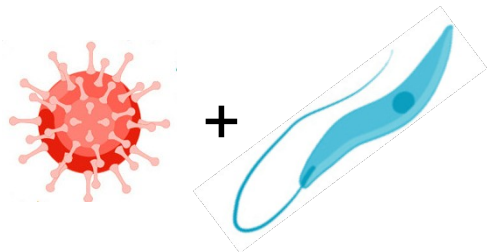
Identify a novel pathogen



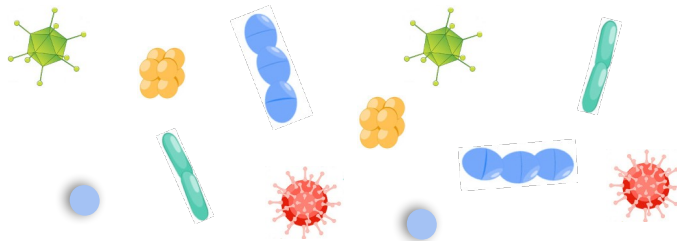
Outbreak detection and prevention



Identify coinfections



Characterize the microbial landscape



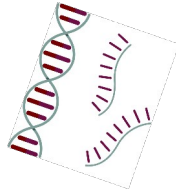
mNGS pipeline

Study design



- Type(s)
- Controls
- Storage
- Lab setup

xNA extraction



- Type(s)
- Controls
- QC

Library prep



- Host removal
- Fragmentation
- dsDNA generation
- End-repair
- Adapter ligation
- Barcoding & PCR
- Clean-up
- Pooling
- QC

Sequencing



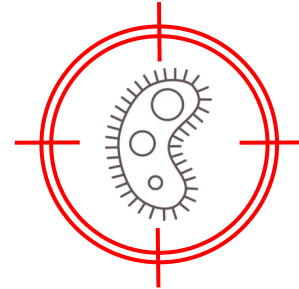
- Machine
- Read length
- Loading conc.

Data analysis



- QC data
- Host removal
- Alignment
- Blast
- Ranking of hits

Pathogen ID



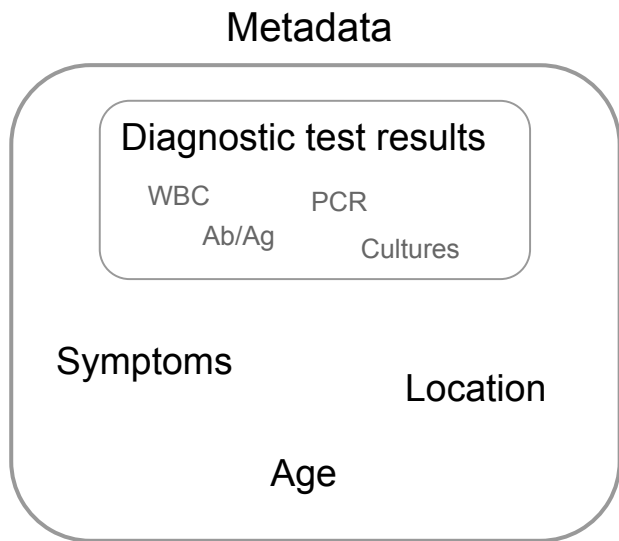
Study design



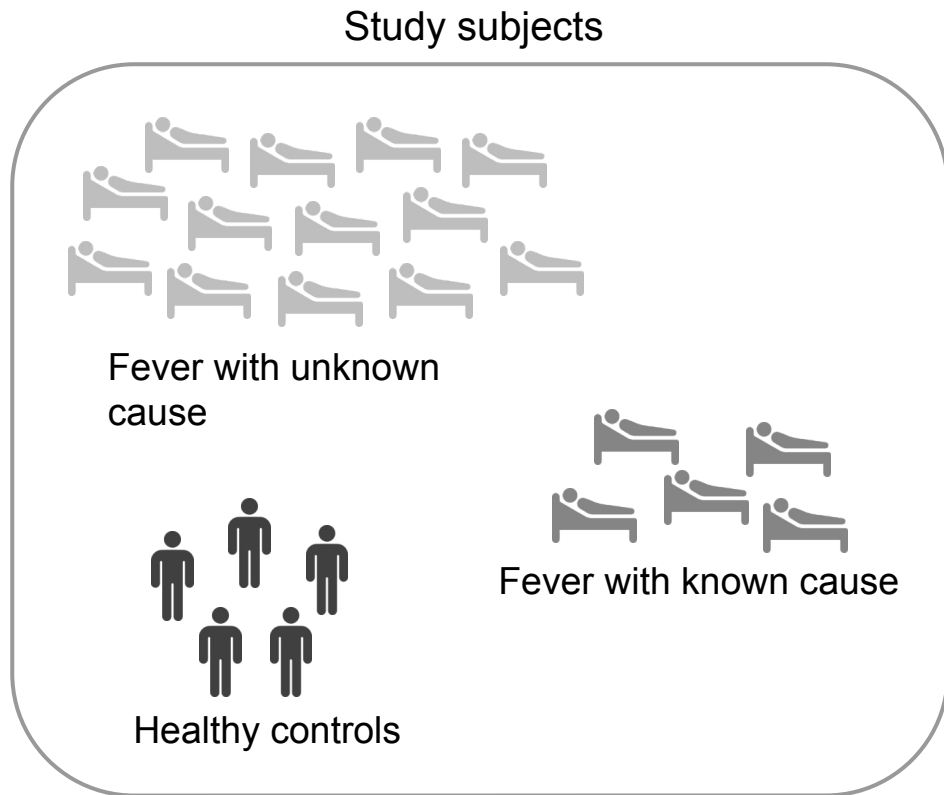


Study design

- Study subjects & metadata



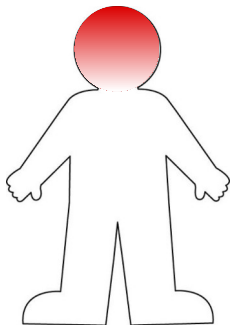
+



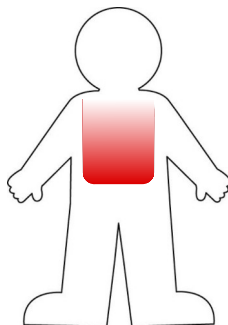


Study design

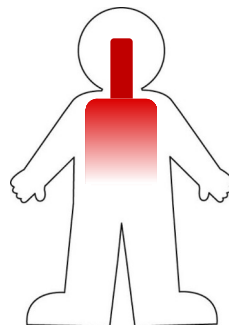
How to choose the sample type? Depends on the question!



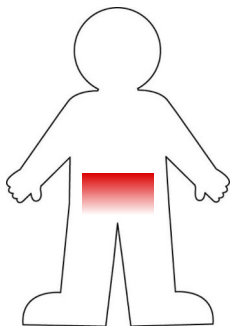
Neurological symptoms?
Meningitis/Encephalitis?
CSF and Serum



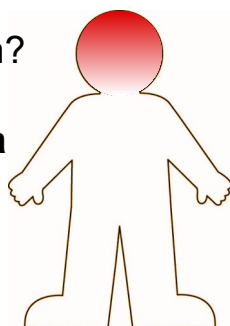
Respiratory symptoms?
Upper = **NP, OP swab, saliva**
Lower= **Bronchial lavage, pleural fluid, lung biopsy**



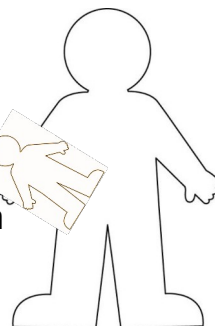
Gastrointestinal symptoms?
Stool, urine, rectal swab, vaginal swab



Systemic infection?
Rash?
Serum or plasma



Neonatal infection?
Paired sample of neonatal CSF/Serum + rectal, vaginal, or serum sample from mother





Study design

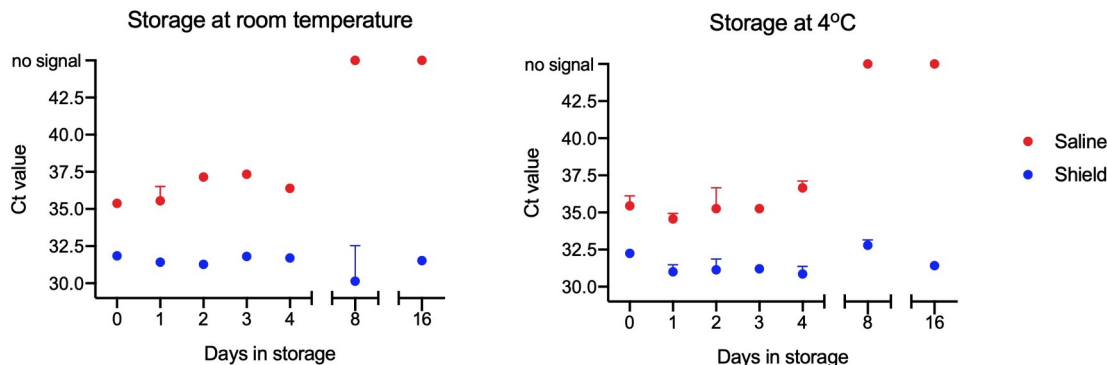
Controls are CRUCIAL to help make sense of your sequencing data

- Healthy controls: clinical samples from healthy controls
 - What are organisms that are commonly found in your population?
- Sampling: mock samples (e.g. needle touching but not piercing skin during CSF sampling)
 - What are the common contaminants that occur during the sampling process?
- During nucleic acid extraction: water controls
 - What are the common contaminants that occur during extraction?
 - What is the level of cross-contamination that occurs between samples during extraction?



Study design

- Sample collection:
 - Collect in nucleic acid stabilizers to preserve sample quality
 - e.g. DNA/RNA Shield (Zymo), RNA/later (Qiagen)
 - Prepare your own collection kits with Shield (cheaper than buying pre-made kits)
 - Collect samples as sterile as possible
 - Collect “collection controls”: e.g. needle touch without aspiration. ...



In vitro transcribed RNA fragments preserve better in Shield than in Saline when stored at room temperature or at 4°C, as assessed by qPCR.



DNA/RNA Shield

- Preserves RNA for 24 hrs at RT, indefinitely at -20°C
- Inactivates pathogens
- 1x Solution for solids & swabs
- 2x Solution for liquids
- -> Order 2x and dilute ½ for swabs if needed



Study design

- Sample storage:
 - Cold chain preserves sample quality



Collect in
DNA/RNA Shield
Or RNA^{later}



Keep cold during
transport



Long term storage
at -80°C



Study design

- Study subjects
 - Population of interest + metadata
 - Healthy controls (!)
- Sample types:
 - Swabs: oral, rectal, vaginal, ...
 - Blood vs serum
 - CSF
- Sample collection:
 - Collect in nucleic acid stabilizers: e.g. DNA/RNA Shield (Zymo), RNA*later* (Qiagen)
 - Work as sterile as possible and collect appropriate controls
- Sample storage:
 - Keep cold during transport (!)
 - Flash freeze as soon as possible (!)

Key points:

- ✓ Preserve sample quality
- ✓ Collect metadata
- ✓ Include controls !
- ✓ Minimize cross-contamination

Example dataset - Fever with unknown etiology

		Known positives						Unknowns						Known negatives						Water	
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	W1	W2
Traditional diagnostics	Malaria blood smear results	+	+	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Human African Trypanosomiasis serology	-	-	-	+	+	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-

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mNGS results	<i>Plasmodium falciparum</i>																				
	<i>Trypanosoma brucei gambiense</i>																				
	<i>Nipah henipavirus</i>																				
	<i>Klebsiella pneumoniae</i>																				
	<i>Pseudomonas aeruginosa</i>																				
	<i>Delftia sp. JD2</i>																				

Example dataset - Fever with unknown etiology

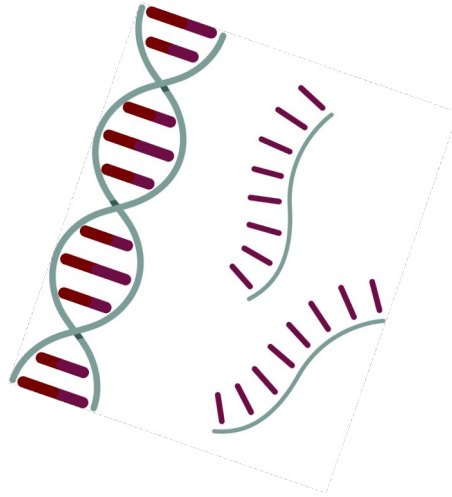
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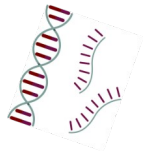
An adequate set of controls will help you discriminate true pathogens from:

- Contaminants during library preparation (extraction to sequencing library)
- Contaminants from sampling procedure

Background signal

xNA extraction





xNA extraction

Key points:

- ✓ Minimize cross-contamination
- ✓ Preserve sample quality

- Before extraction:
 - Spin down to remove host cells (extract supernatant)
- Choose type:
 - DNA:
 - Detect bacteria, parasites, fungi, DNA viruses
 - RNA:
 - All the above + RNA viruses (!)
 - Requires optimal sample quality, degrades very fast
 - Keep cold at all times!
 - TNA (total nucleic acid):
 - DNA + RNA
 - Requires DNase I step for RNA-mNGS
- Column vs bead-based extraction

xNA extraction



Quick-DNA/RNA Microprep Plus

50 preps for \$399

Column-based

Extra reagents to buy:

- ✓ Molecular grade ethanol (200 proof) for washes
- ✓ Isopropanol to add to concentrated wash buffers
- ✓ DNase I included in regular quick-DNA/RNA kits, but not in Pathogen kits
 - Available separately from Qiagen: RNase-Free DNase Set, \$512 for 250 samples

Zymo DNA+RNA extraction kits



Quick-DNA/RNA Miniprep Plus

50 preps for \$399

Column-based



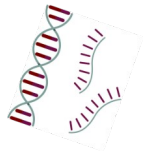
Quick-DNA/RNA Pathogen Magbead

96 preps for \$293

Bead-based

Requires extra:

- ✓ Magnet stand (CZB Engineers)
- ✓ β -mercaptoethanol



xNA extraction

Key point:

✓ xNA quality is crucial

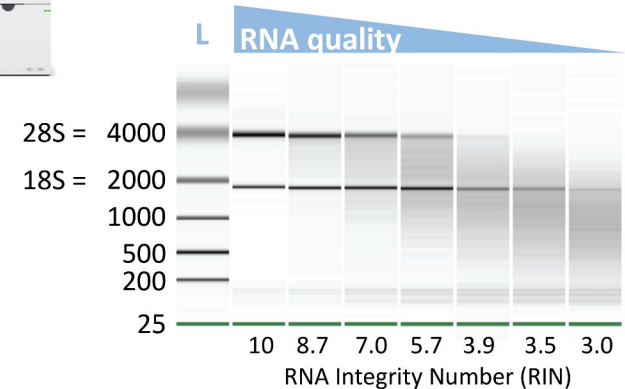
- Quality control:

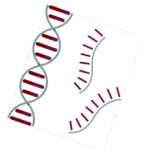
- Quantity (ng/μl):

- Qubit: RNA or DNA High Sensitivity (HS) Assay Kit (ThermoFisher)

- Quality (length):

- Capillary electrophoresis
 - Tapestation (Agilent)
 - High Sensitivity DNA or RNA ScreenTape
 - Gel electrophoresis
 - Regular agarose gel for DNA
 - TBE-urea gel for RNA

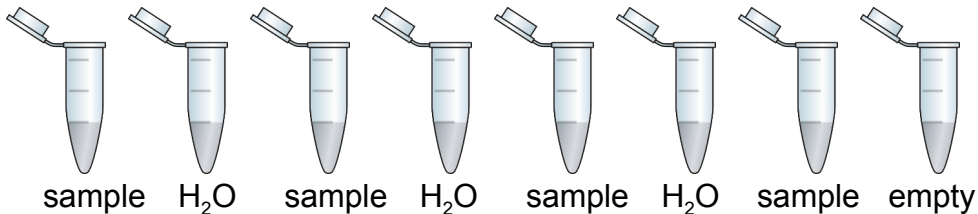




xNA extraction

Keep in mind:

- Magnetic stands for 1.5 ml tubes usually accommodate only 8 tubes at a time
- Cross-contamination should be avoided as much as possible!!
 - Process samples with high pathogen content separately from samples with low pathogen content (if you can)
 - Use water controls to estimate the level of cross-contamination
 - Space out samples with water controls, and/or empty spots in the magnetic rack
 - E.g. for 4 samples:



Next step: Library prep